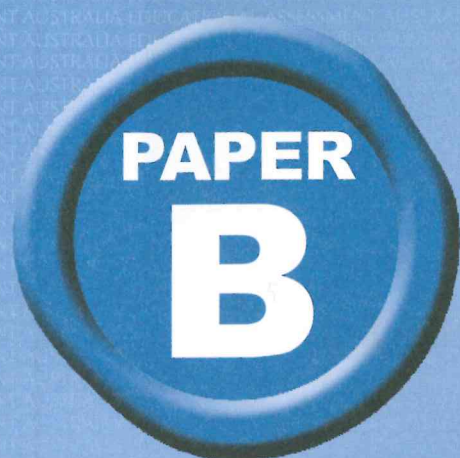




UNSW
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

**EDUCATIONAL ASSESSMENT
AUSTRALIA**

Year 4



ICAS

**INTERNATIONAL COMPETITIONS AND
ASSESSMENTS FOR SCHOOLS**

**SCIENCE
2009**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

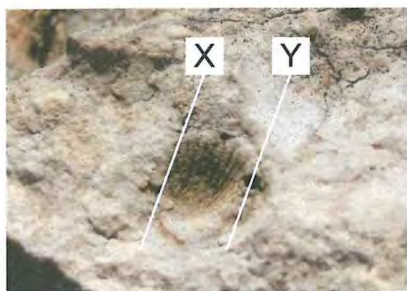
Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

**PLEASE SEE BACK COVER FOR A LIST
OF THE YEAR LEVELS THAT SHOULD
SIT THIS PAPER**

1. The photograph shows the actual size of a fossil of a shell.



What is the approximate width of the fossil from X to Y?

- (A) 1 cm
- (B) 2 cm
- (C) 3 cm
- (D) 4 cm

2. Sarah looked at some floor coverings in her house.

Floor covering	Hard	Soft	Rough	Smooth
tiles	✓			✓
grass matting		✓	✓	
wooden boards	✓		✓	
carpet		✓		✓

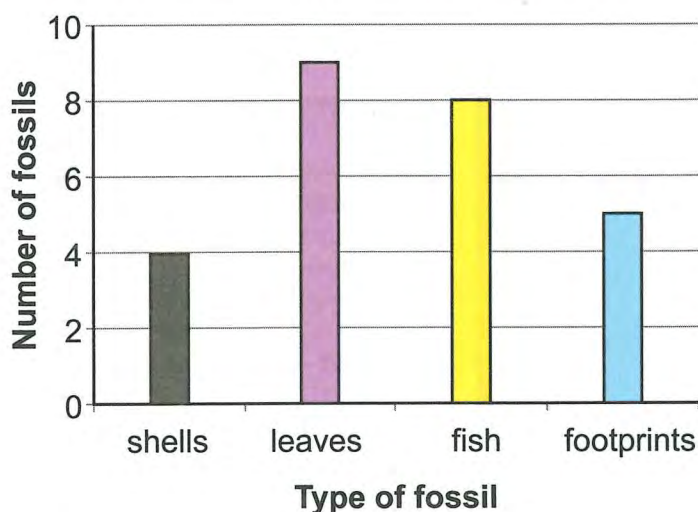
Hard, smooth surfaces are slippery when wet.

Which floor covering in Sarah's house is slippery when wet?

- (A) tiles
- (B) grass matting
- (C) wooden boards
- (D) carpet

3. A geologist found different types of fossils while investigating a dry river bed.

The graph shows the number of each type of fossil she found.



How many fish fossils did she find?

- (A) 4
- (B) 5
- (C) 8
- (D) 9

4. The photograph shows the habitat of an animal.

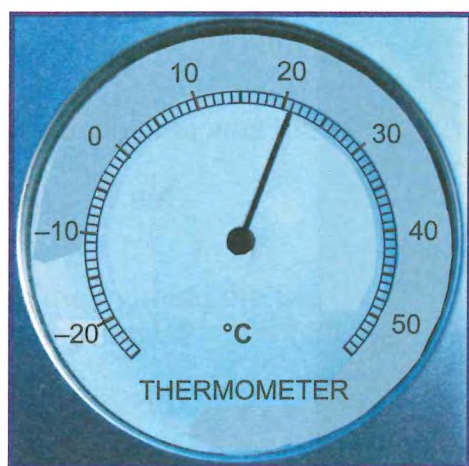


Which animal can be seen in this habitat?

- (A) bird
- (B) worm
- (C) spider
- (D) caterpillar

5. A thermometer measures temperature in degrees Celsius ($^{\circ}\text{C}$).

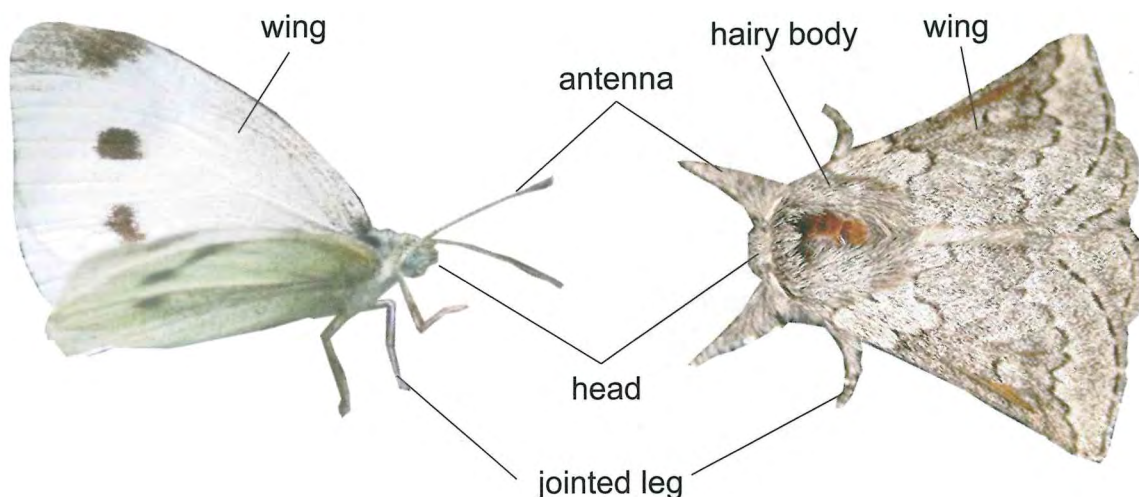
The photograph shows a thermometer.



What temperature ($^{\circ}\text{C}$) is shown on the thermometer?

- (A) 22
- (B) 21
- (C) 20
- (D) 19

6. The photographs show a butterfly and a moth at rest. The butterfly is viewed from the side. The moth is viewed from above. Some body parts are labelled.



Which feature is common to both animals?

- (A) hairy body
 - (B) long thin antennae
 - (C) legs that are jointed
 - (D) position of wings when resting
7. The photograph shows a rock being measured with a centimetre ruler.



What is the approximate length of this rock?

- (A) 3 cm
- (B) 6 cm
- (C) 9 cm
- (D) 12 cm

8. Schist and gneiss are two types of metamorphic rock that have layers.

- Schist has layers that are black and shiny.
- Gneiss has light and dark layers.

The photographs show some small rocks.



W



X



Y



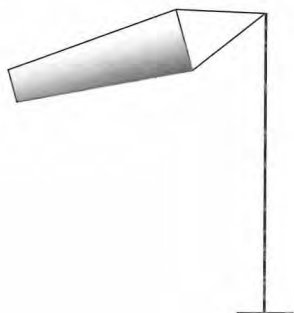
Z

Which two rocks are most likely to be gneiss?

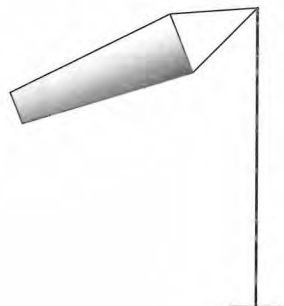
- (A) W and X (B) W and Y (C) X and Y (D) X and Z

For questions 9 and 10 use the information below.

The diagram shows a windsock at four different times during a day.



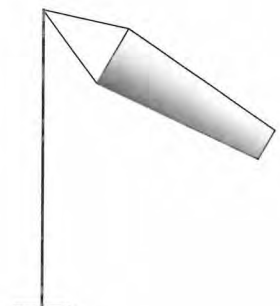
6 am



12 noon



6 pm



12 midnight

9. At which time was the wind the strongest?

- (A) 6 am (B) 12 noon (C) 6 pm (D) 12 midnight

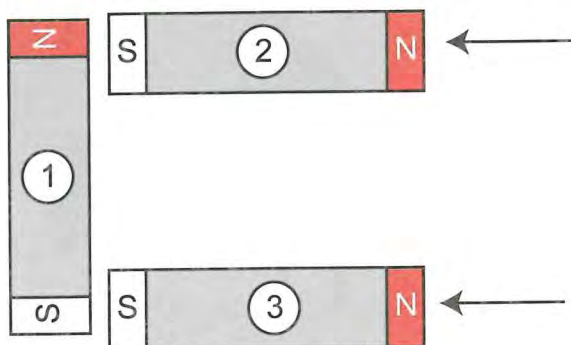
10. In the morning the wind was coming from the west. In which direction was the windsock being blown?

- (A) north (B) west (C) south (D) east

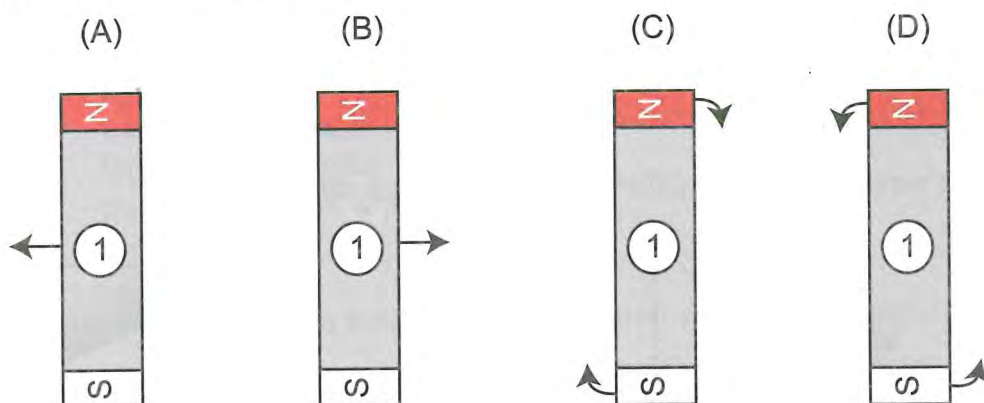
11. A magnet has a south pole and a north pole.

A south pole will move away from the south pole of another magnet. A south pole will move towards the north pole of another magnet.

The drawing shows two magnets, magnet 2 and magnet 3, being moved towards magnet 1.



How will magnet 1 move?



12. Adam wanted to test the accuracy of the weekly weather forecast printed in his local newspaper.

He recorded the minimum and maximum temperature every day for one week, and compared this data with what had been forecast for that week.

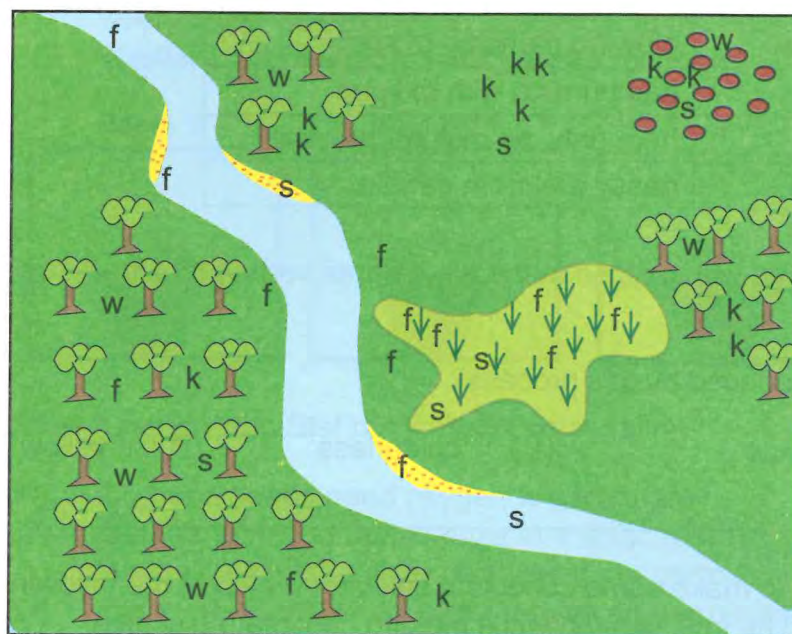
What was the purpose of Adam's actions?

He wanted to find

- (A) the minimum and maximum temperature for the week.
- (B) how close the actual conditions were to the forecast.
- (C) the range in temperature over the week.
- (D) the average weekly temperature.

13. Jai drew a map of his local area to show where

- particular environments exist
- individual animals have been observed.

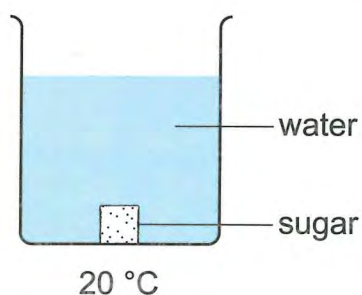


KEY	
Environments	
	grasslands
	swamp lands
	rocky area
	river
	wood lands
	sandy beach
Animals	
f	frog
k	kangaroo
s	snake
w	wombat

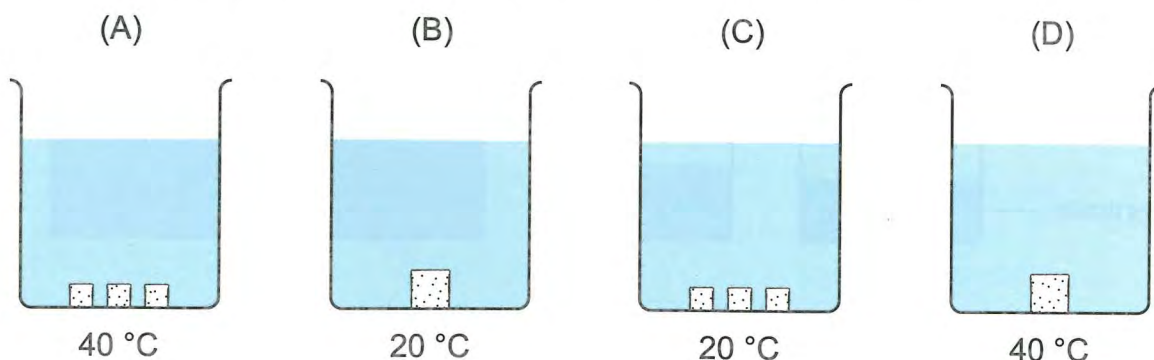
Which animal has been observed in all six environments in Jai's local area?

- (A) frog (B) snake (C) wombat (D) kangaroo

14. Linda wanted to know if sugar dissolves faster in warm water than in cold water. This is how she set up the first part of her experiment.



Which set-up should Linda use to correctly complete her experiment?



15. The photograph shows the zones that occur in the flame of a burning candle.

The table gives information about these zones.



Zone	Colour of light produced	Light is produced by
1	yellow	glowing particles of soot
2	red	fuel burning with oxygen
3	-	heat only, no oxygen present
4	blue	gassy fuel from candle

When fuel from a candle combines with oxygen in the air, it forms carbon dioxide and water.

What is the colour of the light produced by this process?

- (A) red (B) blue (C) colourless (D) yellow

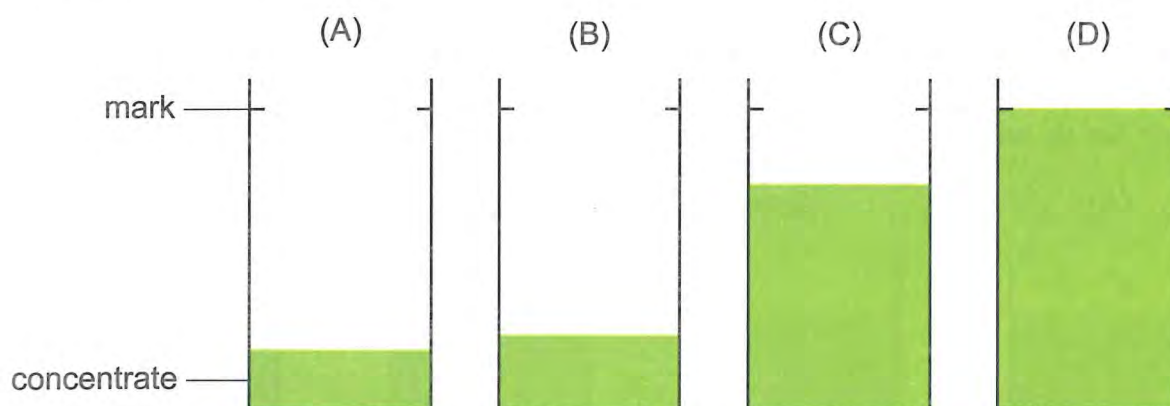
16. Mr Robinson's class is planning to make some chocolate biscuits. Which of the following is a scientific question that could be investigated while making chocolate biscuits?

- (A) Do chocolate biscuits taste nice?
(B) Which type of biscuits does the class prefer?
(C) Do chocolate biscuits need baking powder to rise?
(D) How much does it cost to make chocolate biscuits?

17. Jenny bought a bottle of fruit juice concentrate. The instructions said that she should mix 1 part concentrate to 4 parts water.

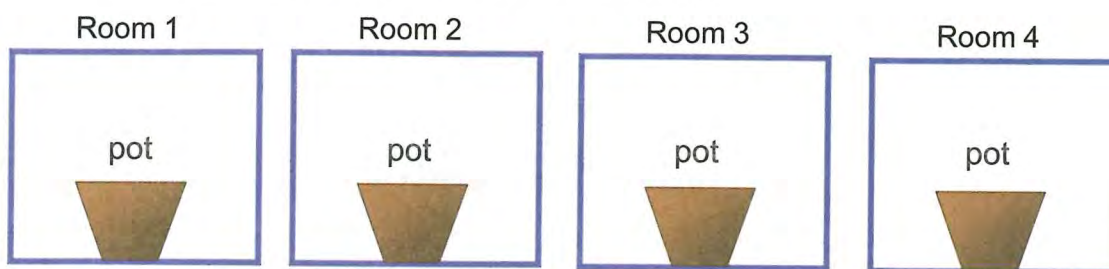
She wanted to make one container of fruit juice.

Which container shows the correct amount of concentrate that she should use, before filling to the mark with water?



18. Stef set up an experiment with the following conditions:

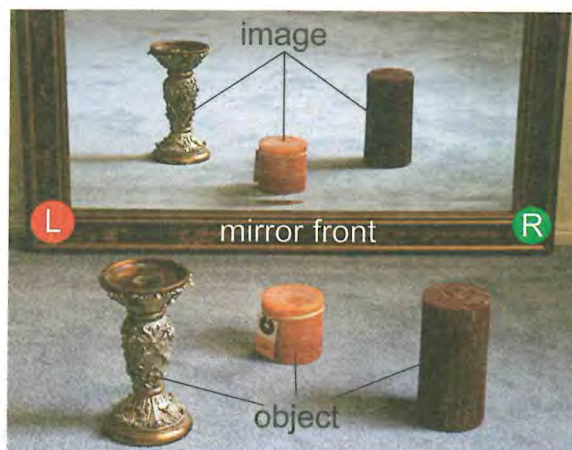
- each pot was the same size
- 10 bean seeds were planted in each pot
- the soil type in each pot was the same
- each pot was given the same amount of water
- every room was the same size and had the same amount and direction of sunlight
- a different temperature was set in each room.



What question was Stef most likely investigating?

- (A) Do bean seeds need oxygen to germinate?
- (B) Can germinating beans grow in a small room?
- (C) What is the best soil for bean seeds to germinate?
- (D) What effect does temperature have on germinating bean seeds?

19. Sue took this photograph of some objects and their images in a mirror.



What is true about the images?

- (A) Larger objects have smaller images.
- (B) The images have different shapes to the objects.
- (C) The images show the same side of the objects as seen directly by Sue.
- (D) Objects closer to the mirror have images that appear closer to the mirror.

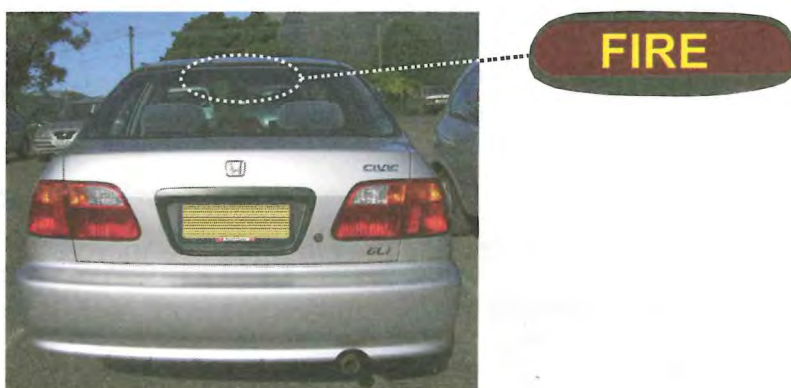
20. Khalid wanted to find out which of the four laundry liquids he had was better for removing cooking oil stains. He carried out the following steps:

- He dipped a large piece of cloth into cooking oil and then cut the oil-stained cloth into four equal pieces.
- He mixed each laundry liquid with warm water.
- He used each mixture to scrub a piece of oil-stained cloth.

What conditions should Khalid keep the same to have a fair test?

- (A) type of laundry liquid, type of oil, amount of stain left
- (B) water temperature, amount of laundry liquid, scrubbing time
- (C) amount of laundry liquid, scrubbing time, amount of stain left
- (D) water temperature, amount of laundry liquid, type of laundry liquid

21. A fire truck has a sign that appears as "FIRE" in the rear vision mirror of a car that is in front of the fire truck.



Which option shows the sign on the front of the fire truck?

(A)



(B)



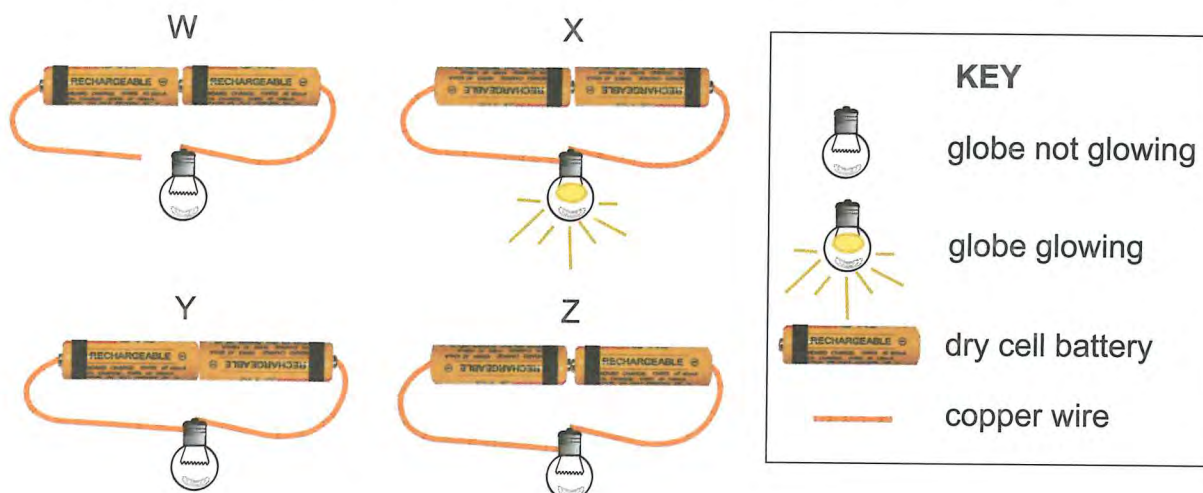
(C)



(D)



22. Vanessa was investigating how she could make an electrical circuit using two batteries, some wire and a globe. She used four different arrangements, W, X, Y and Z.

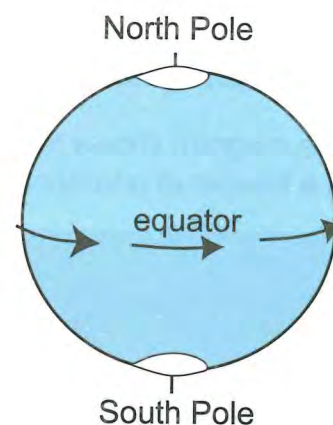


Only arrangement X made the globe light up.

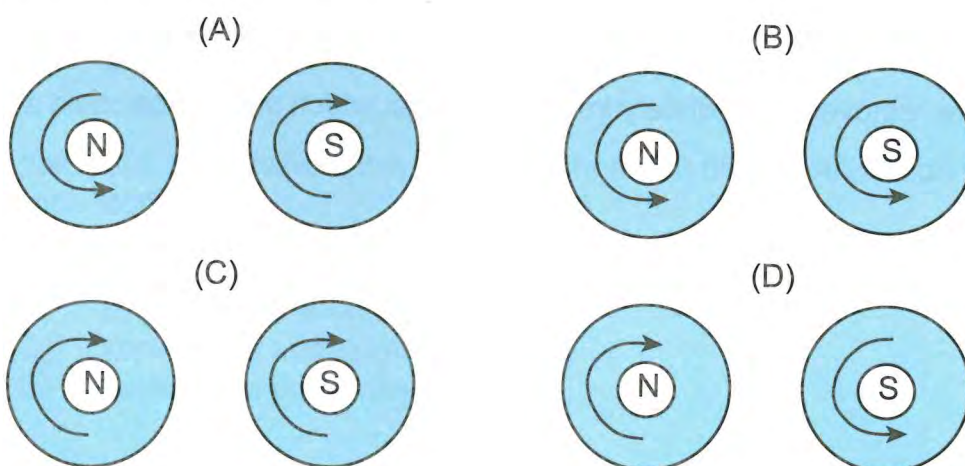
She concluded that the globe will glow only when the batteries are facing in

- (A) the same direction in the circuit and the circuit is open.
- (B) the same direction in the circuit and the circuit is closed.
- (C) opposite directions in the circuit and the circuit is open.
- (D) opposite directions in the circuit and the circuit is closed.

23. The drawing shows how the Earth rotates when looking from above the equator.

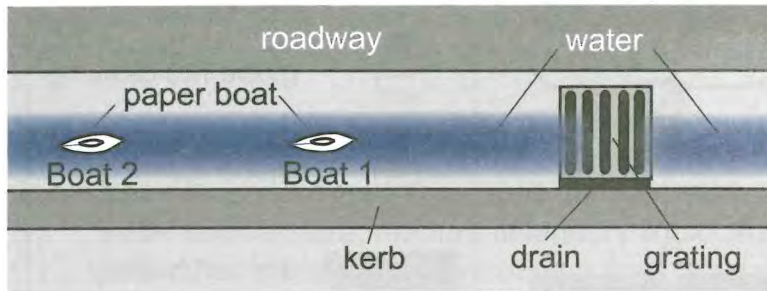


Which two drawings show how the Earth rotates when looking from above the North Pole and when looking from above the South Pole?



24. Luka floated two identical paper boats on water in a gutter. The water was flowing towards a drain.

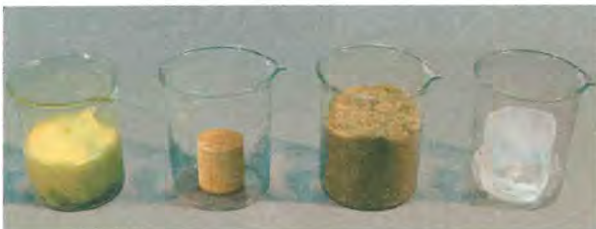
The drawing shows the position of the boats at a particular moment in time.



Boat 1 went down the drain 10 seconds after this.

What most likely happened to Boat 2?

- (A) It remained in the same position.
(B) It went down the drain about 5 seconds after Boat 1.
(C) It went down the drain about 10 seconds after Boat 1.
(D) It went down the drain about 20 seconds after Boat 1.
25. Solids have a fixed volume and hold their shape.
Liquids have a fixed volume and take the shape of any container into which they are placed.
- Liquids and gases are fluids. Fluids spill or flow away if they are not contained.
- The diagram shows four containers of different substances. The substances were all kept in a freezer at a temperature which was below 0 °C.



butter wood sand ice

If the containers were removed immediately from the freezer, which substance could act as a fluid?

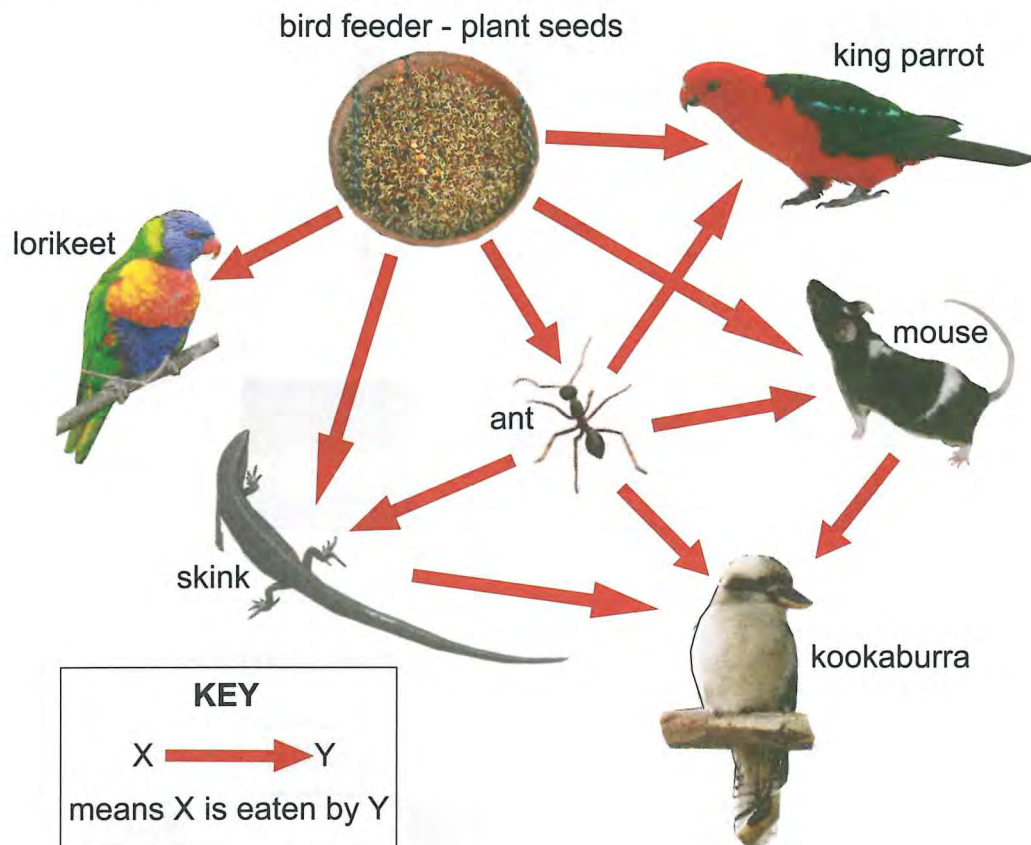
- (A) butter
(B) wood
(C) sand
(D) ice

For questions 26 and 27 use the information below.

There are different types of animals.

- Animals that only eat plant materials are called herbivores.
- Animals that only eat other animals are called carnivores.
- Animals that eat both plant materials and other animals are called omnivores.

The diagram shows the food web created when animals were attracted to a garden by a bird feeder.



26. Which of the animals are omnivores?

- (A) kookaburra only
- (B) ant, lorikeet only
- (C) king parrot, mouse, skink only
- (D) ant, king parrot, kookaburra, mouse, skink only

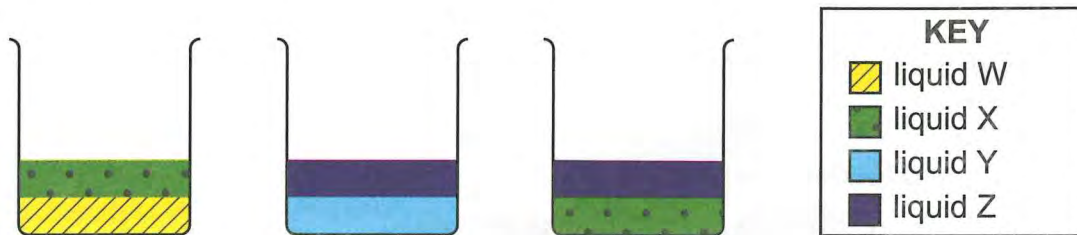
27. Which two animals from this food web are **not** in direct competition with each other for the same food?

- (A) ant and skink
- (B) king parrot and skink
- (C) kookaburra and skink
- (D) kookaburra and lorikeet

28. Misha has four different liquids, W, X, Y and Z.

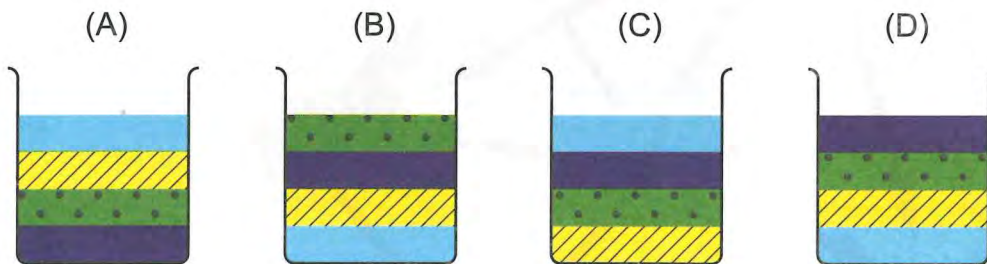
When the liquids are poured together they do not mix but form separate layers.

The diagrams show what happened when Misha made three different combinations from these liquids.

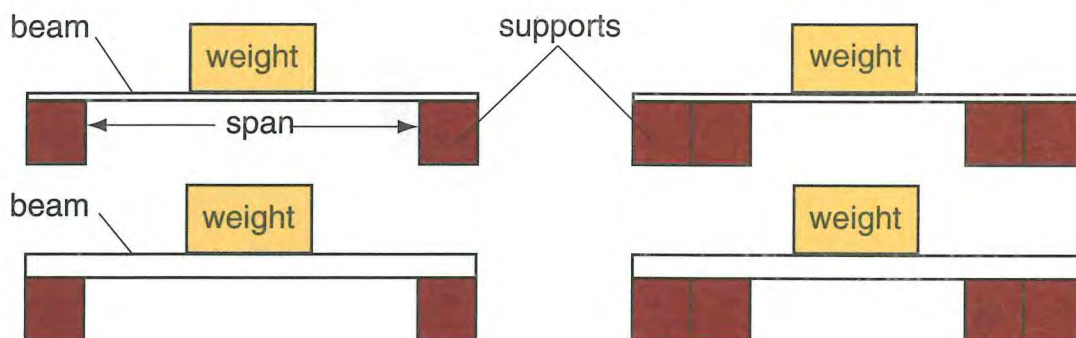


Afterwards, Misha combined all four different liquids.

Which of the following could show Misha's results?



29. Grace built the four model bridges shown and investigated how strong they were.



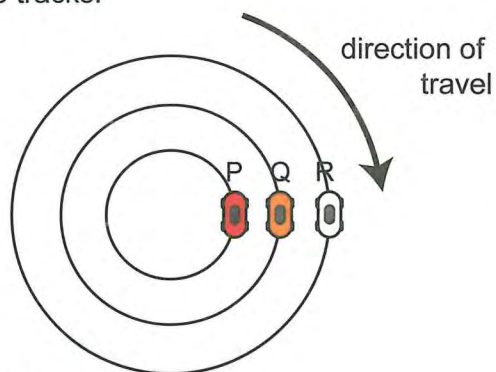
What two features did Grace vary to see how they affected bridge strength?

- (A) the height of the beam and the length of the span
- (B) the thickness of the beam and the length of the span
- (C) the thickness of the beam and the height of the beam
- (D) the size of the weight and the width of the supports

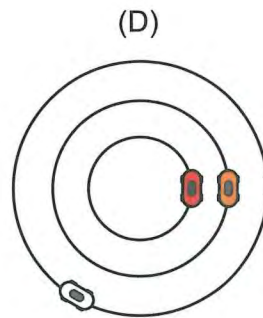
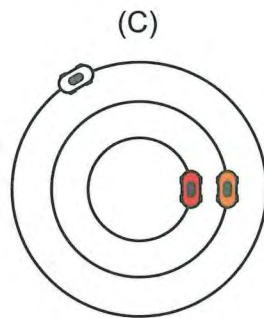
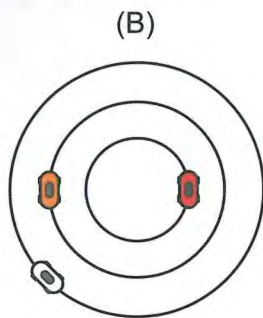
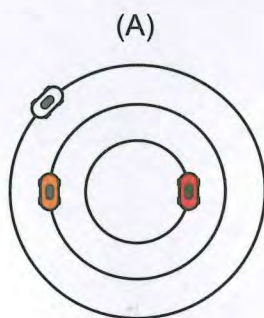
30. Three toy trains P, Q and R are travelling around three separate tracks. All three trains are travelling at the same speed.

Train P takes 1 minute to completely travel around its track.
 Train Q takes 2 minutes to completely travel around its track.
 Train R takes 3 minutes to completely travel around its track.

At certain times the three trains are level with each other as shown.



Which diagram shows the position of the trains five minutes later?



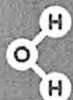
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AUSTRALIA: Year 4
BRUNEI: Primary 4
INDONESIA: Year 5
MALAYSIA: Standard 4
NEW ZEALAND: Year 5
PACIFIC: Year 4
SINGAPORE: Primary 3
SOUTH AFRICA: Grade 4

**PAPER
B**



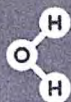
SC

2009 Science Answer Keys

ICAS
International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Paper I	Paper J
1	D	A	C	A	C	A	A	B	D	D
2	D	A	B	C	B	B	A	A	D	D
3	C	C	C	A	A	A	C	C	B	B
4	C	D	D	B	D	A	C	B	D	D
5	D	B	B	B	A	D	C	D	C	C
6	A	C	D	B	B	A	C	B	C	C
7	B	A	B	C	D	C	A	C	B	B
8	D	B	A	D	B	B	C	B	B	B
9	C	A	A	B	A	D	A	B	B	B
10	A	D	A	A	B	B	B	B	B	B
11	A	C	D	D	D	A	B	C	C	C
12	C	B	C	A	A	A	D	A	D	D
13	D	B	D	C	B	C	D	C	A	A
14	C	D	C	C	A	C	B	D	D	D
15	C	A	B	D	A	B	B	B	C	C
16	B	C	A	B	C	C	B	D	D	D
17	A	A	A	D	C	D	D	A	C	C
18	B	D	B	D	A	D	B	A	C	C
19	B	D	D	B	C	D	A	A	A	A
20	A	B	B	A	D	C	A	B	A	A
21	D	D	A	D	D	B	C	B	C	C
22	D	B	B	B	C	C	B	C	A	A
23	D	A	C	C	C	D	C	C	A	A
24	C	C	A	D	C	C	C	A	B	B
25	B	C	C	C	C	B	A	C	A	A
26	A	C	B	B	C	A	B	A	A	A
27	A	D	D	D	D	C	C	D	D	D
28	B	D	C	C	A	C	D	C	C	C
29	C	B	D	C	D	B	B	A	B	B
30	B	A	D	B	B	B	D	B	A	A

(Please turn over)



SC

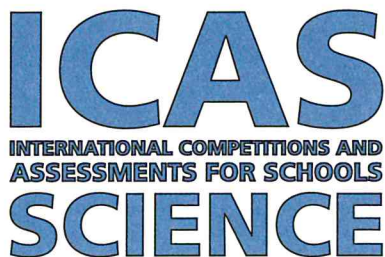
2009 Science Answer Keys

ICAS
International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Paper I	Paper J
31			D	B	A	A	C	D	B	B
32			C	C	C	D	A	D	B	B
33			B	A	D	B	B	D	B	B
34			C	D	D	D	B	B	D	D
35			A	B	B	D	A	D	C	C
36			D	A	B	A	B	A	D	D
37			B	A	B	B	B	D	D	D
38			A	D	B	C	D	C	D	D
39			C	A	D	D	D	D	C	C
40			D	B	D	D	A	A	B	B
41					C	C	D	A	A	A
42					B	C	B	D	C	C
43					B	D	C	C	B	B
44					D	D	D	C	B	B
45					C	C	D	B	D	D

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Web: www.eaa.unsw.edu.au

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**EDUCATIONAL ASSESSMENT
AUSTRALIA**

- Rub out all mistakes completely.
- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.

FIRST NAME

D	E	B	B	I	E
○	○	○	○	○	○
A	A	A	A	A	A
B	B	●	●	B	○
C	C	C	C	C	○
D	D	D	D	D	○

LAST NAME

B	A	C	H
○	○	○	○
A	●	A	A
●	B	B	○
C	C	●	○
D	D	D	○

FIRST NAME **LAST NAME**

C	H	A	N
A	A	A	A
B	B	B	B
C	C	C	C
D	D	D	D

A	I	B	E	N	G		
A	A	A	A	A	A	A	A
B	B	B	B	B	B	B	B
C	C	C	C	C	C	C	C
D	D	D	D	D	D	D	D

FIRST NAME										LAST NAME			
J	A	M	A	L		B	I	N		A	B	A	S
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[illegible][illegible]

☐ Male ☐ Female



☐ Yes ☐ No

Postcode: _____

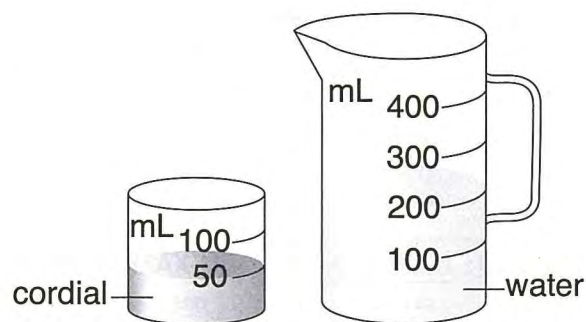
Day		Month		Year	
0	0	0	0	0	0
1	1	1	1	1	1
2	2		2	2	2
3	3		3	3	3
	4		4	4	4
	5		5	5	5
	6		6	6	6
	7		7	7	7
	8		8	8	8
	9		9	9	9

	(A)	(K)
	(B)	(L)
	(C)	(M)
	(D)	(N)
	(E)	(O)
	(F)	(P)
	(G)	(Q)
	(H)	(R)
	(I)	(S)
	(J)	(T)

TO ANSWER THE QUESTIONS

Example: Ari added cordial to water to make a jug of drink.
What will be the volume of the drink in the jug?

- (A) 50 mL
- (B) 150 mL
- (C) 200 mL
- (D) 250 mL



The answer is 250 mL, so you would fill in the oval **D**, as shown.

A **B** **C** **D**



START

- 1 **A** **B** **C** **D**
- 2 **A** **B** **C** **D**
- 3 **A** **B** **C** **D**
- 4 **A** **B** **C** **D**
- 5 **A** **B** **C** **D**
- 6 **A** **B** **C** **D**
- 7 **A** **B** **C** **D**
- 8 **A** **B** **C** **D**
- 9 **A** **B** **C** **D**
- 10 **A** **B** **C** **D**
- 11 **A** **B** **C** **D**
- 12 **A** **B** **C** **D**
- 13 **A** **B** **C** **D**
- 14 **A** **B** **C** **D**
- 15 **A** **B** **C** **D**

- 16 **A** **B** **C** **D**
- 17 **A** **B** **C** **D**
- 18 **A** **B** **C** **D**
- 19 **A** **B** **C** **D**
- 20 **A** **B** **C** **D**
- 21 **A** **B** **C** **D**
- 22 **A** **B** **C** **D**
- 23 **A** **B** **C** **D**
- 24 **A** **B** **C** **D**
- 25 **A** **B** **C** **D**
- 26 **A** **B** **C** **D**
- 27 **A** **B** **C** **D**
- 28 **A** **B** **C** **D**
- 29 **A** **B** **C** **D**
- 30 **A** **B** **C** **D**

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